THE KUNNETH FORMULA FOR THE COHOMOLOGY OF A PRODUCT OF QUASI-COMPACT SCHEMES WITH AFFINE DIAGONAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Vanishing of higher cohomology of quasi-coherent modules on affine schemes. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-quasi-coherent-affine-cohomology-zero (tag 01XB). Supplies the affine case of the theorem.
- Functoriality of Cech cohomology and Cech-to-cohomology comparison. Stacks Project, Cohomology, Lemma cohomology-lemma-functoriality-cech (tag 01EO). Produces the maps between Cech complexes over the three coverings used in the proof.

Source reference. Original lemma 0BEF: sources/stacks-chow/tags.txt, source lines 14348–14348. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01FD: sources/stacks-chow/tags.txt, source lines 1866–1866. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07MB: sources/stacks-chow/tags.txt, source lines 9721–9721. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0BDX: sources/stacks-chow/tags.txt, source lines 14330–14330. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XD: sources/stacks-chow/tags.txt, source lines 2514–2514. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 0BDY: sources/stacks-chow/tags.txt, source lines 14331–14331. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01FF: sources/stacks-chow/tags.txt, source lines 1868–1868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01F4: sources/stacks-chow/tags.txt, source lines 1857–1857. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X and Y be schemes over k and let $\mathcal{F}$, resp. $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_X$-module, resp. $\mathcal{O}_Y$-module. Then we have a canonical isomorphism*

$$H^n(X \times_{\mathrm{Spec}(k)} Y, pr_1^*\mathcal{F} \otimes_{\mathcal{O}_{X\times_{\mathrm{Spec}(k)}Y}} pr_2^*\mathcal{G}) = \bigoplus\nolimits_{p+q=n} H^p(X,\mathcal{F}) \otimes_k H^q(Y,\mathcal{G})$$

provided X and Y are quasi-compact and have affine diagonal[1] *(for example if X and Y are separated).*

Proof. In this proof unadorned products and tensor products are over k. As maps

$$H^p(X,\mathcal{F}) \otimes H^q(Y,\mathcal{G}) \longrightarrow H^n(X \times Y, \mathrm{pr}_1^*\mathcal{F} \otimes_{\mathcal{O}_{X\times Y}} \mathrm{pr}_2^*\mathcal{G})$$

we use functoriality of cohomology to get maps $H^p(X,\mathcal{F}) \to H^p(X \times Y, \mathrm{pr}_1^*\mathcal{F})$ and $H^p(Y,\mathcal{G}) \to H^p(X \times Y, \mathrm{pr}_2^*\mathcal{G})$ and then we use the cup product

$$\cup : H^p(X \times Y, \mathrm{pr}_1^*\mathcal{F}) \otimes H^q(X \times Y, \mathrm{pr}_2^*\mathcal{G}) \longrightarrow H^n(X \times Y, \mathrm{pr}_1^*\mathcal{F} \otimes_{\mathcal{O}_{X\times Y}} \mathrm{pr}_2^*\mathcal{G})$$

The result is true when X and Y are affine by the vanishing of higher cohomology groups on affines (Cohomology of Schemes, Lemma 01XB) and the definitions (of pullbacks of quasi-coherent modules and tensor products of quasi-coherent modules).

Choose finite affine open coverings $\mathcal{U} : X = \bigcup_{i\in I} U_i$ and $\mathcal{V} : Y = \bigcup_{j\in J} V_j$. This determines an affine open covering $\mathcal{W} : X \times Y = \bigcup_{(i,j)\in I\times J} U_i \times V_j$. Note that $\mathcal{W}$ is a refinement of $\mathrm{pr}_1^{-1}\mathcal{U}$ and of $\mathrm{pr}_2^{-1}\mathcal{V}$. Thus by Cohomology, Lemma 01FD we obtain maps

$$\check{\mathcal{C}}^\bullet(\mathcal{U},\mathcal{F}) \to \check{\mathcal{C}}^\bullet(\mathcal{W}, \mathrm{pr}_1^*\mathcal{F}) \quad \text{and} \quad \check{\mathcal{C}}^\bullet(\mathcal{V},\mathcal{G}) \to \check{\mathcal{C}}^\bullet(\mathcal{W}, \mathrm{pr}_2^*\mathcal{G})$$

compatible with pullback maps on cohomology. In Cohomology, Equation (07MB) we have constructed a map of complexes

$$\mathrm{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{W}, \mathrm{pr}_1^*\mathcal{F}) \otimes \check{\mathcal{C}}^\bullet(\mathcal{W}, \mathrm{pr}_2^*\mathcal{G})) \longrightarrow \check{\mathcal{C}}^\bullet(\mathcal{W}, \mathrm{pr}_1^*\mathcal{F} \otimes_{\mathcal{O}_{X\times Y}} \mathrm{pr}_2^*\mathcal{G})$$

defining the cup product on cohomology. Combining the above we obtain a map of complexes

$$\mathrm{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{U},\mathcal{F}) \otimes \check{\mathcal{C}}^\bullet(\mathcal{V},\mathcal{G})) \longrightarrow \check{\mathcal{C}}^\bullet(\mathcal{W}, \mathrm{pr}_1^*\mathcal{F} \otimes_{\mathcal{O}_{X\times Y}} \mathrm{pr}_2^*\mathcal{G}) \tag{0.1.1}$$

We warn the reader that this map is not an isomorphism of complexes. Recall that we may compute the cohomologies of our quasi-coherent sheaves using our coverings (Cohomology of Schemes, Lemmas 0BDX and 01XD). Thus on cohomology (0.1.1) reproduces the map of the lemma.

Consider a short exact sequence $0 \to \mathcal{F} \to \mathcal{F}' \to \mathcal{F}'' \to 0$ of quasi-coherent modules. Since the construction of (0.1.1) is functorial in $\mathcal{F}$ and since the formation of the

[1] The case where X and Y are quasi-separated will be discussed in Lemma 0BEF below.

relevant Čech complexes is exact in the variable $\mathcal{F}$ (because we are taking sections over affine opens) we find a map between short exact sequence of complexes

$$\begin{array}{ccccc} \mathrm{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{U},\mathcal{F})\otimes\check{\mathcal{C}}^\bullet(\mathcal{V},\mathcal{G})) & \longrightarrow & \mathrm{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{U},\mathcal{F}')\otimes\check{\mathcal{C}}^\bullet(\mathcal{V},\mathcal{G})) & \longrightarrow & \mathrm{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{U},\mathcal{F}'')\otimes\check{\mathcal{C}}^\bullet(\mathcal{V},\mathcal{G})) \\ \downarrow & & \downarrow & & \downarrow \\ \check{\mathcal{C}}^\bullet(\mathcal{W},\mathrm{pr}_1^*\mathcal{F}\otimes_{\mathcal{O}_{X\times Y}}\mathrm{pr}_2^*\mathcal{G}) & \longrightarrow & \check{\mathcal{C}}^\bullet(\mathcal{W},\mathrm{pr}_1^*\mathcal{F}'\otimes_{\mathcal{O}_{X\times Y}}\mathrm{pr}_2^*\mathcal{G}) & \longrightarrow & \check{\mathcal{C}}^\bullet(\mathcal{W},\mathrm{pr}_1^*\mathcal{F}''\otimes_{\mathcal{O}_{X\times Y}}\mathrm{pr}_2^*\mathcal{G}) \end{array}$$

(we have dropped the outer zeros). Looking at long exact cohomology sequences we find that if the result of the lemma holds for 2-out-of-3 of $\mathcal{F},\mathcal{F}',\mathcal{F}''$, then it holds for the third.

Observe that X has finite cohomological dimension for quasi-coherent modules, see Cohomology of Schemes, Lemma 01XI. Using induction on $d(\mathcal{F}) = \max\{d \mid H^d(X,\mathcal{F}) \neq 0\}$ we will reduce to the case $d(\mathcal{F}) = 0$. Assume $d(\mathcal{F}) > 0$. By Cohomology of Schemes, Lemma 0BDY we have seen that there exists an embedding $\mathcal{F} \to \mathcal{F}'$ such that $H^p(X,\mathcal{F}') = 0$ for all $p \geq 1$. Setting $\mathcal{F}'' = \mathrm{Coker}(\mathcal{F} \to \mathcal{F}')$ we see that $d(\mathcal{F}'') < d(\mathcal{F})$. Then we can apply the result from the previous paragraph to see that it suffices to prove the lemma for $\mathcal{F}'$ and $\mathcal{F}''$ thereby proving the induction step.

Arguing in the same fashion for $\mathcal{G}$ we find that we may assume that both $\mathcal{F}$ and $\mathcal{G}$ have nonzero cohomology only in degree 0. Let $V \subset Y$ be an affine open. Consider the affine open covering $\mathcal{U}_V : X \times V = \bigcup_{i\in I} U_i \times V$. It is immediate that

$$\check{\mathcal{C}}^\bullet(\mathcal{U},\mathcal{F})\otimes\mathcal{G}(V) = \check{\mathcal{C}}^\bullet(\mathcal{U}_V,\mathrm{pr}_1^*\mathcal{F}\otimes_{\mathcal{O}_{X\times Y}}\mathrm{pr}_2^*\mathcal{G})$$

(equality of complexes). We conclude that

$$R\mathrm{pr}_{2,*}(\mathrm{pr}_1^*\mathcal{F}\otimes_{\mathcal{O}_{X\times Y}}\mathrm{pr}_2^*\mathcal{G}) \cong \Gamma(X,\mathcal{F})\otimes_k\mathcal{G} \cong \bigoplus\nolimits_{\alpha\in A}\mathcal{G}$$

on Y. Here A is a basis for the k-vector space $\Gamma(X,\mathcal{F})$. Cohomology on Y commutes with direct sums (Cohomology, Lemma 01FF). Using the Leray spectral sequence for pr_2 (via Cohomology, Lemma 01F4) we conclude that $H^n(X\times Y,\mathrm{pr}_1^*\mathcal{F}\otimes_{\mathcal{O}_{X\times Y}}\mathrm{pr}_2^*\mathcal{G})$ is zero for $n > 0$ and isomorphic to $H^0(X,\mathcal{F})\otimes H^0(Y,\mathcal{G})$ for $n = 0$. This finishes the proof (except that we should check that the isomorphism is indeed given by cup product in degree 0; we omit the verification). □